Trade Data Quality Report

Imports & Exports — Jan 2024 to Sep 2025

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Executive Summary

PURPOSE: Exploratory analysis to identify data quality issues and recommend cleaning actions for Brazil-Canada trade report generation.

IMPORT DATASET: 63,716 valid records

- Zero Quantity: 27,481 (43.1%)
- Outliers: 4,235
- Multi-UOM Products: 0
- Duplicates: 0 (0.0%)

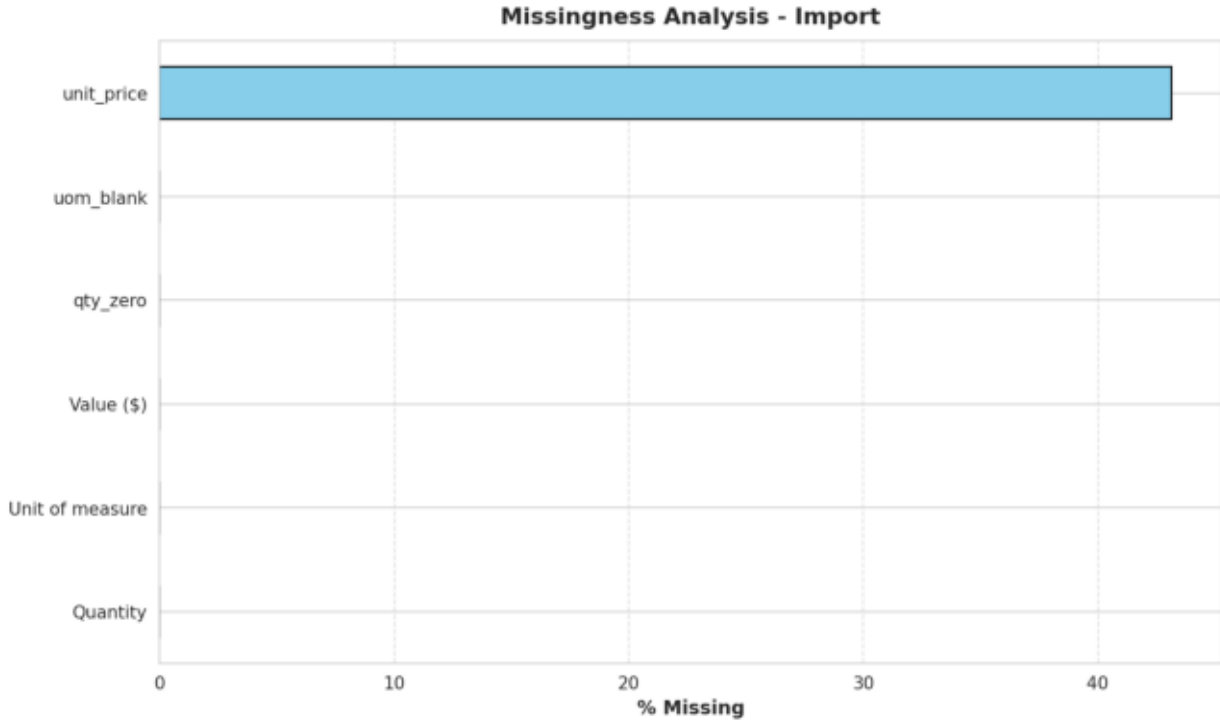
EXPORT DATASET: 14,964 valid records

- Zero Quantity: 9,498 (63.5%)
- Outliers: 788
- Multi-UOM Products: 2
- Duplicates: 0 (0.0%)

CRITICAL CLEANING NEEDS:

1. Fix Period datatype (currently object, should be datetime)
2. Drop 100% empty columns (State)
3. Handle zero quantities (impute or flag)
4. Standardize units of measure per product
5. Flag or winsorize price outliers by category

Missingness Analysis - Import



Key Findings:

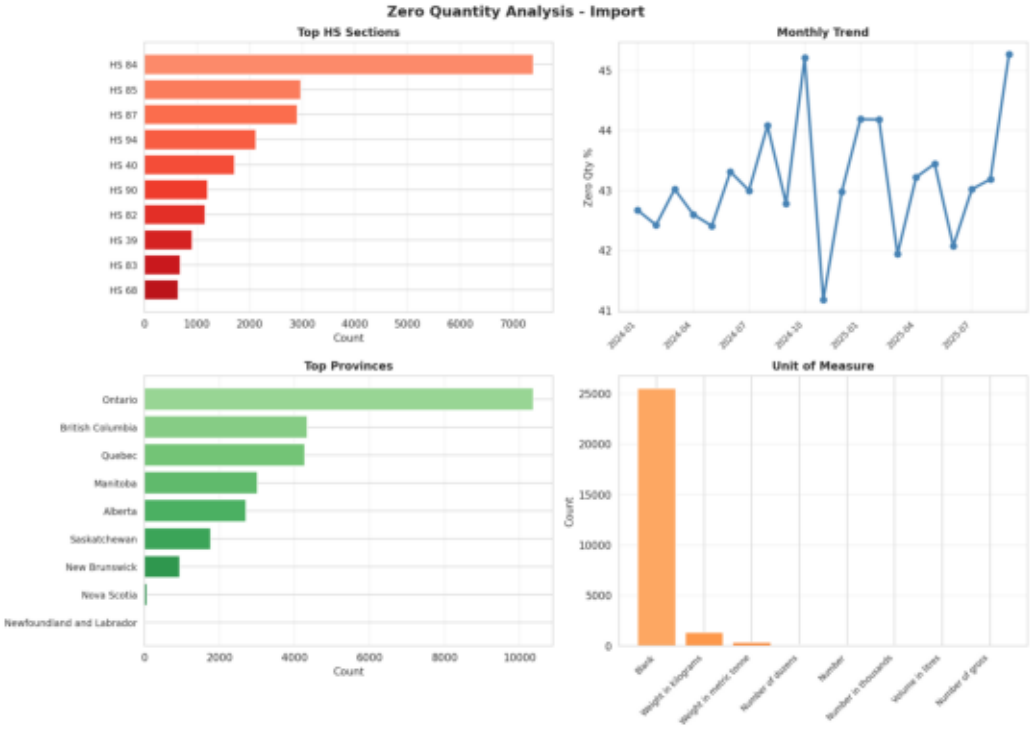
- Unit price missing in 43.1% of records
- Missing unit prices caused by Quantity=0 (division by zero)
- Core fields have minimal missing data

Data Cleaning Actions:

→ Impute Quantity from Value using category-specific median unit_price

→ OR flag zero-quantity records and exclude from price calculations

Zero Quantity Analysis - Import



Key Findings:

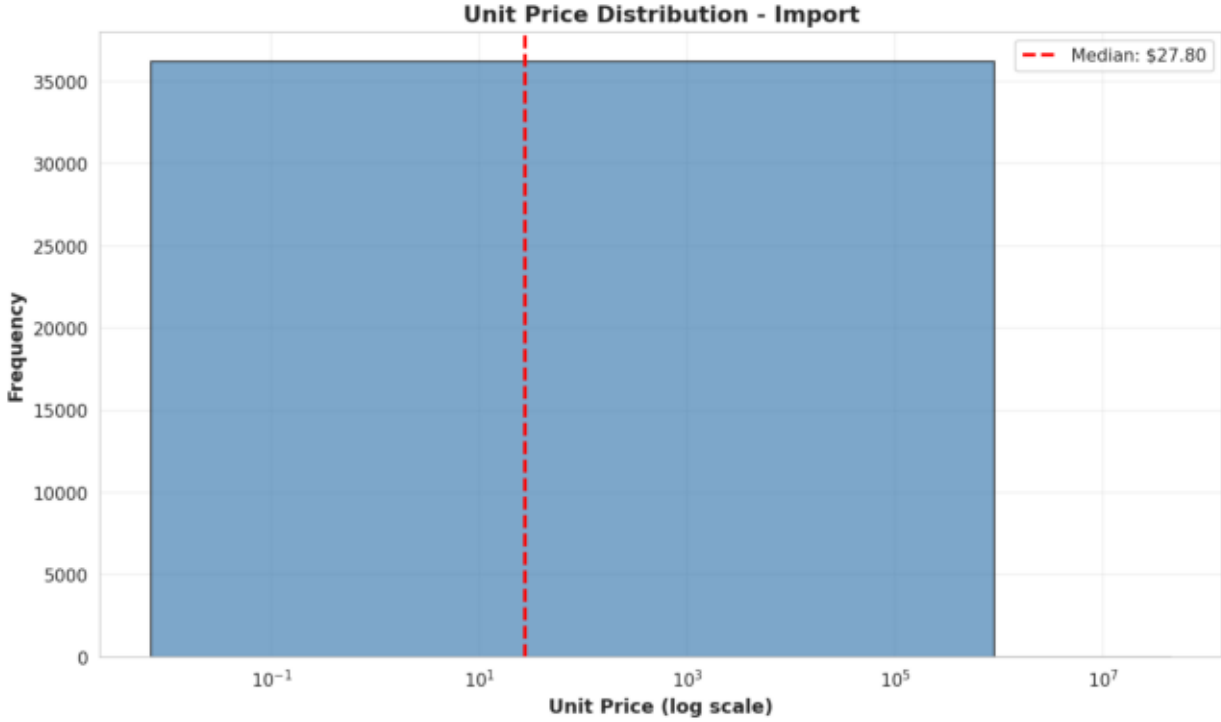
- Zero quantity affects 43.1% of records have Quantity=0
- Pattern is consistent across time periods
- Certain HS sections show higher concentration

Data Cleaning Actions:

→ Create 'qty_imputed' flag column for tracking

→ Impute quantity OR separate these records into audit file

Unit Price Distribution - Import



Key Findings:

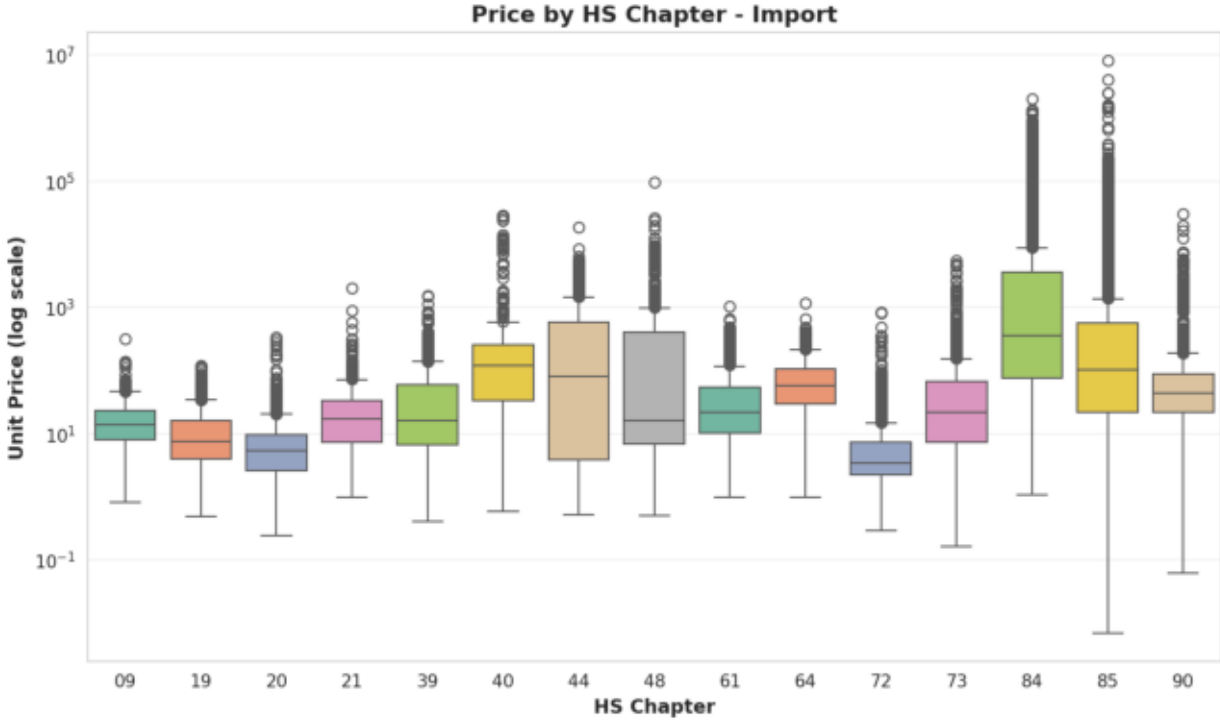
- Prices span 7+ orders of magnitude
- Most prices between $10 - 1000$
- Long tail of expensive items

Data Cleaning Actions:

→ Log-transform prices before statistical modeling

→ Create price_category field for segmentation

Price by HS Chapter - Import



Key Findings:

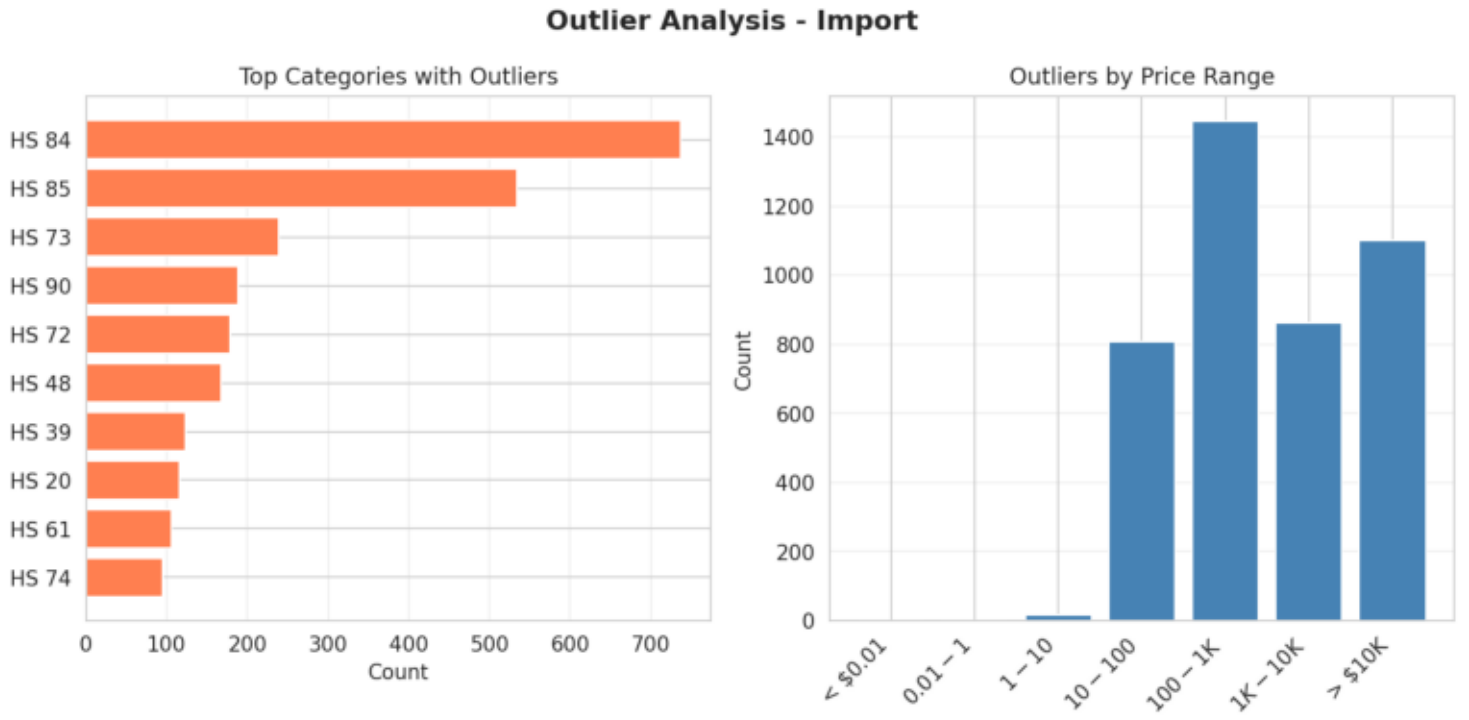
- Each HS chapter has distinct price distribution
- Machinery/electronics show higher medians
- Within-category variance is high

Data Cleaning Actions:

→ Calculate outliers WITHIN each HS2 category, not globally

→ Use category-specific thresholds

Outlier Analysis - Import



Key Findings:

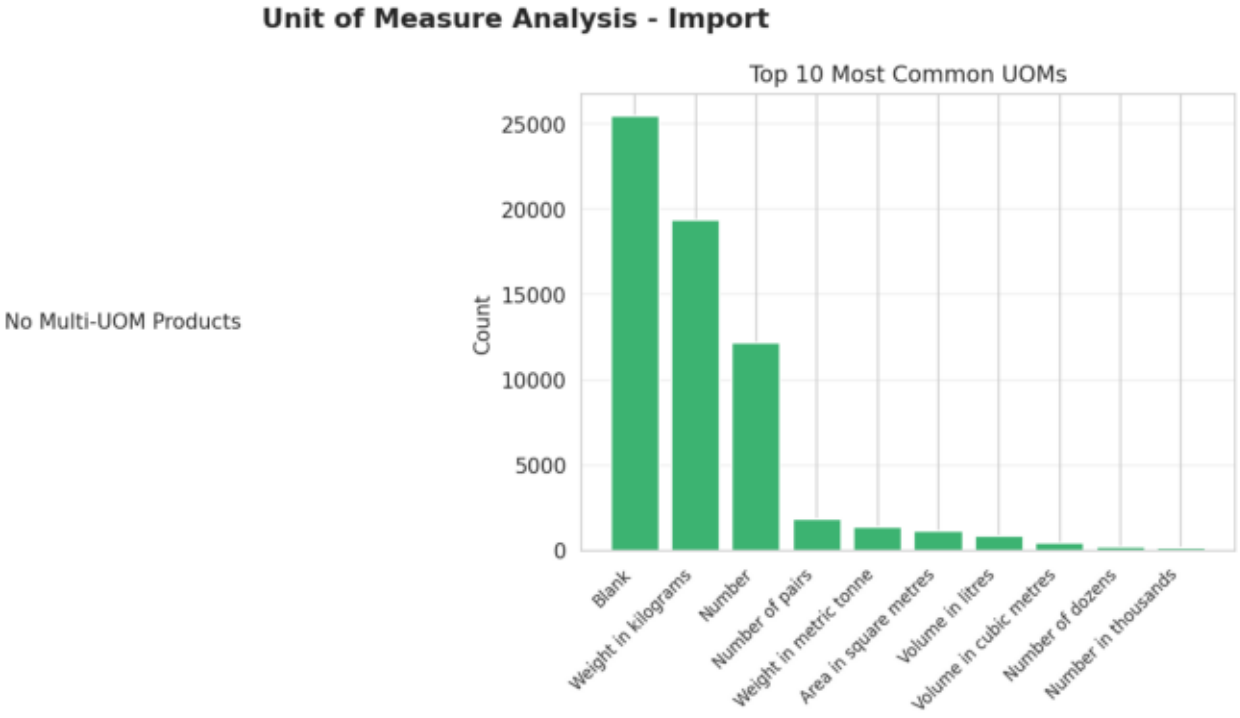
- 4,235 outliers detected using category-specific IQR
- Most are extreme high values
- Distributed across multiple HS sections

Data Cleaning Actions:

→ Create 'is_outlier' boolean column for filtering

→ Winsorize at Q1-3*IQR / Q3+3*IQR OR exclude from aggregations

UOM Analysis - Import



Key Findings:

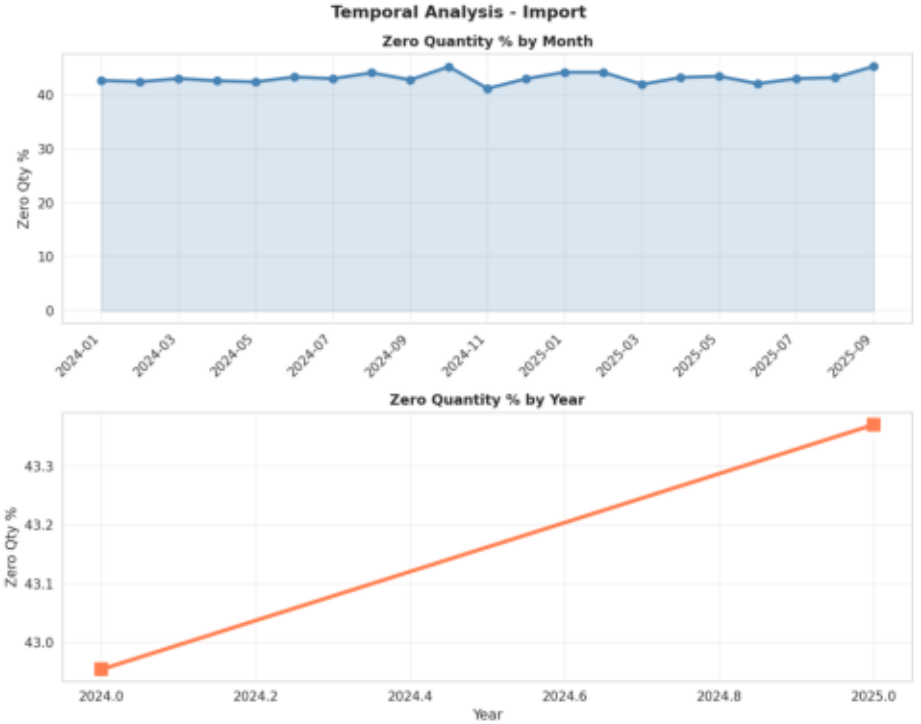
- 0 products use multiple units
- Cannot sum quantities without standardization
- Weight-based units are most common

Data Cleaning Actions:

→ Standardize to ONE unit per HS code

→ Build conversion table and apply before aggregation

Temporal Analysis - Import



Key Findings:

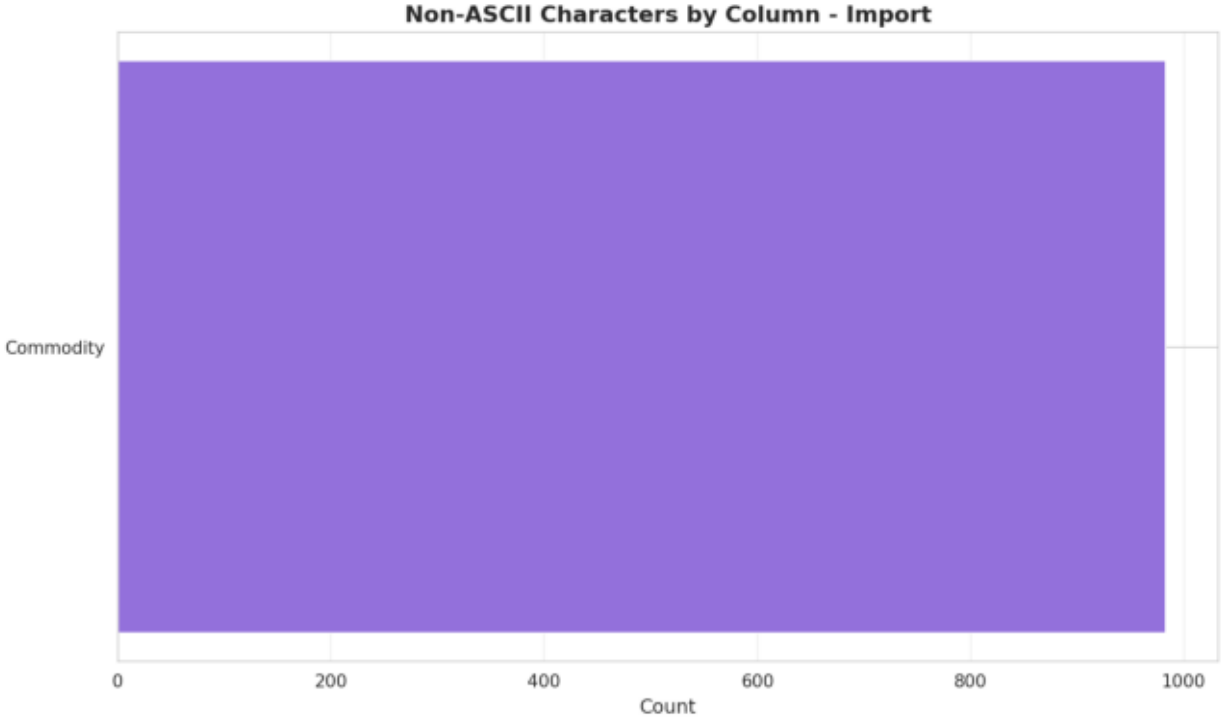
- Trade patterns show expected seasonal variation
- No major anomalies visible
- Zero-quantity rate is stable over time

Data Cleaning Actions:

→ Keep temporal structure intact for time-series analysis

→ Add month/quarter columns for aggregation

Non-ASCII Characters - Import



Key Findings:

• 983 records with special characters

• Mainly in descriptions and location names

• Could cause export or matching issues

Data Cleaning Actions:

→ Normalize text using unidecode library

→ Apply to Commodity, Province, Country columns

Missingness Analysis - Export

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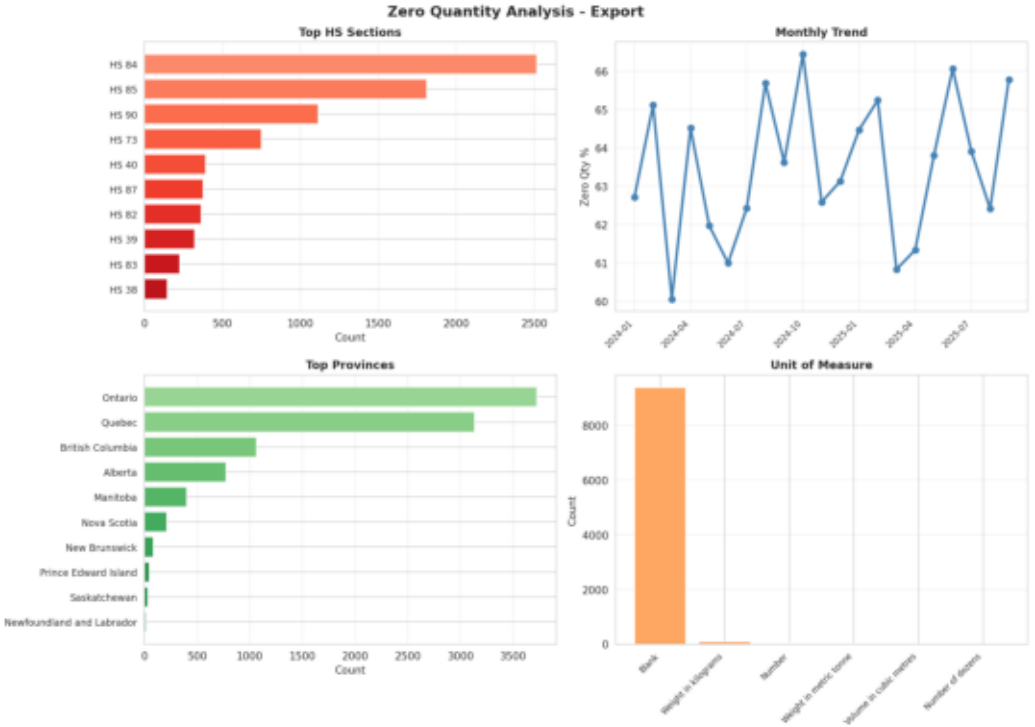

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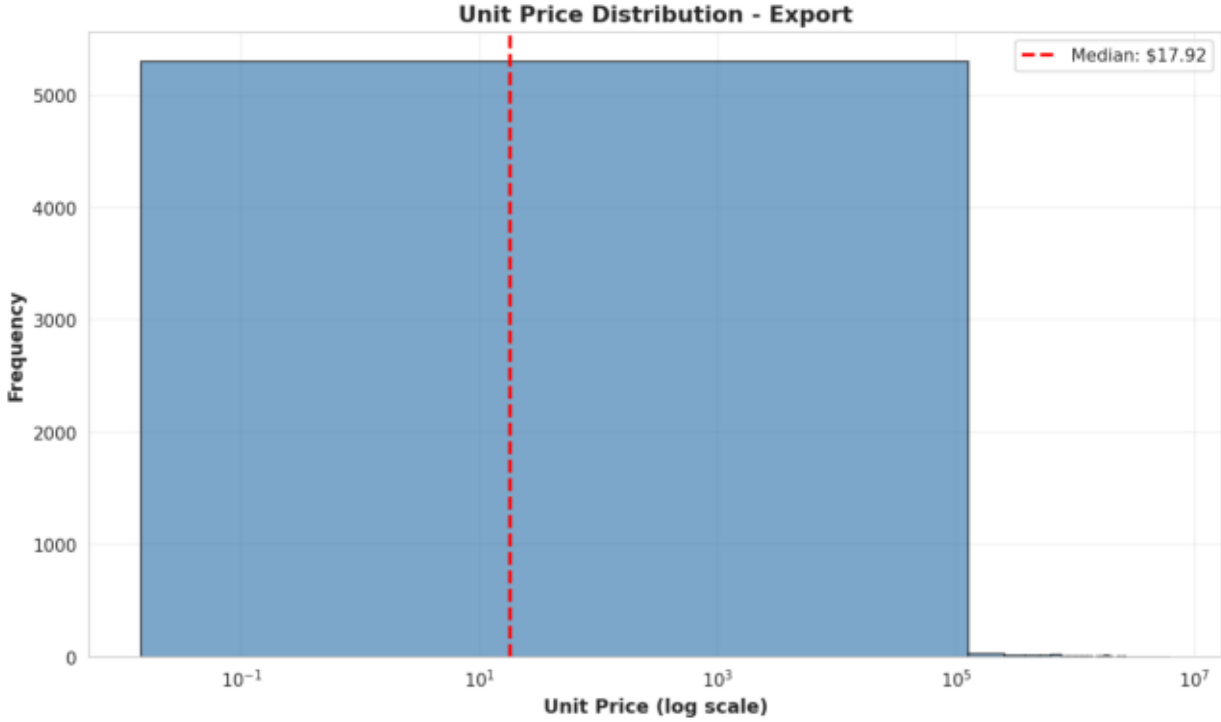

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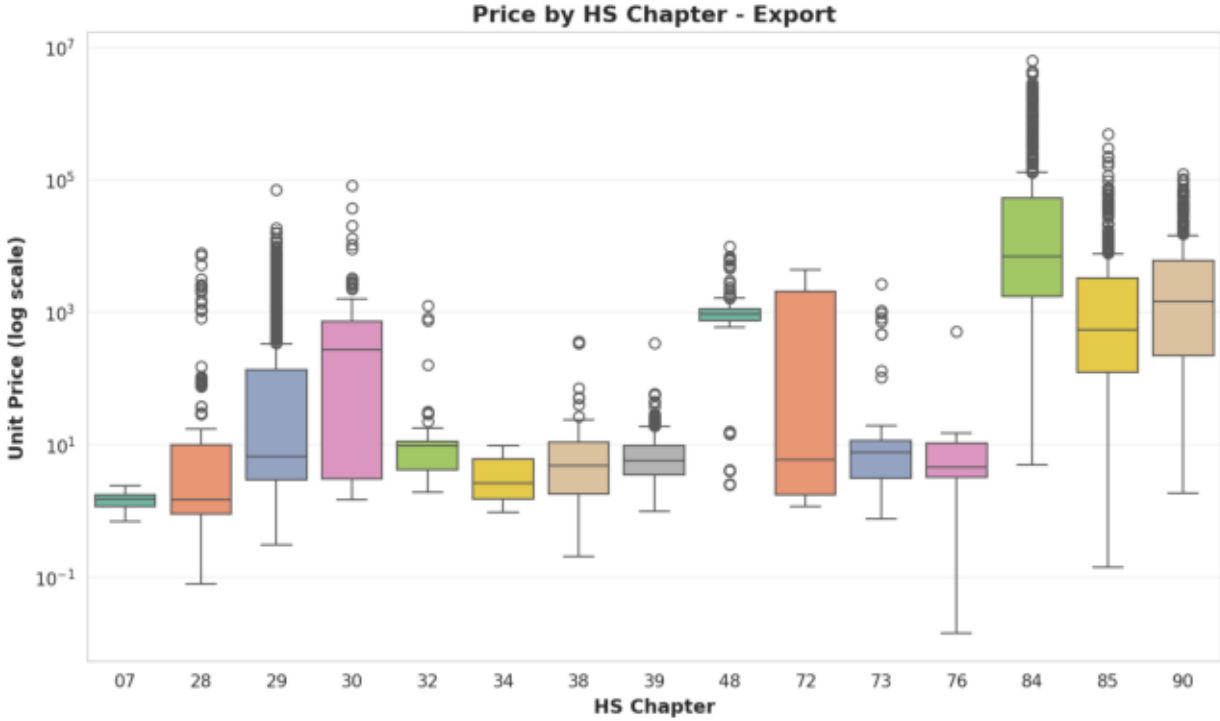

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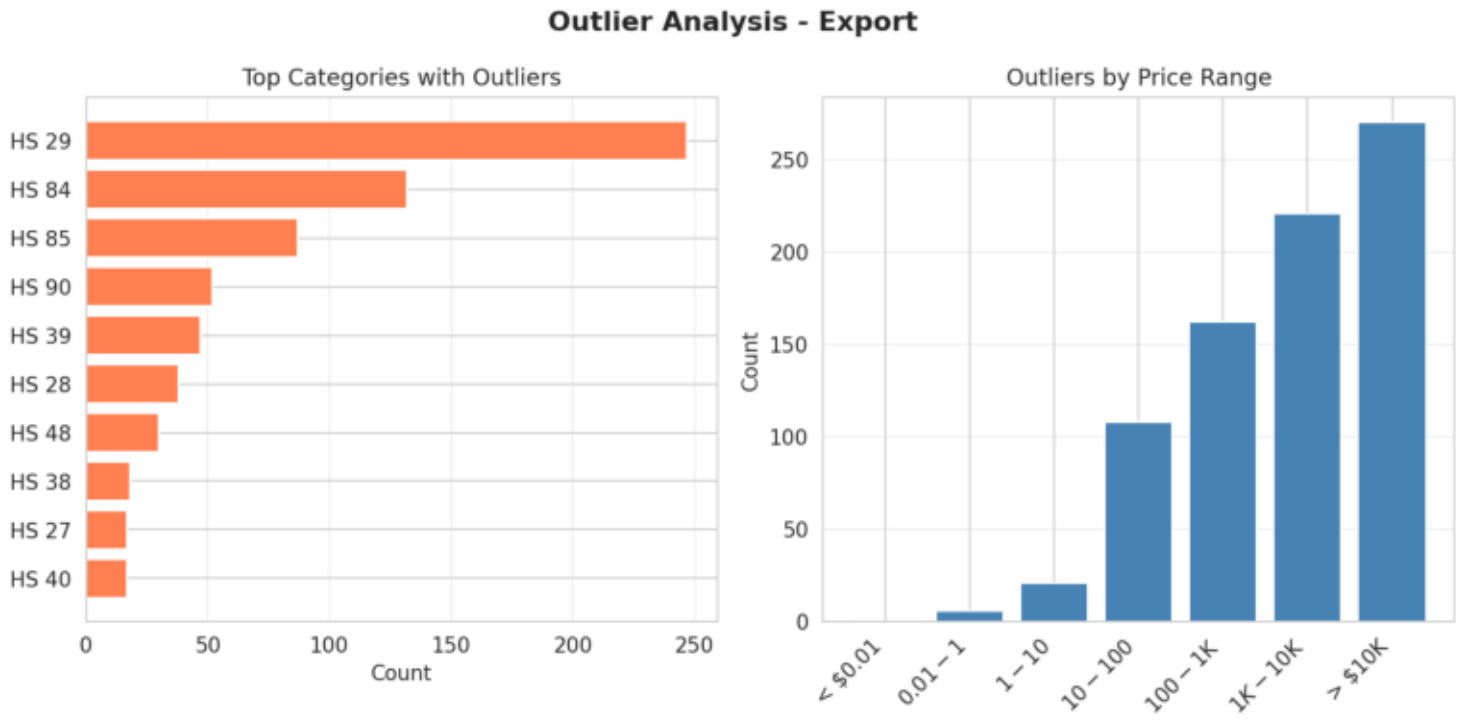

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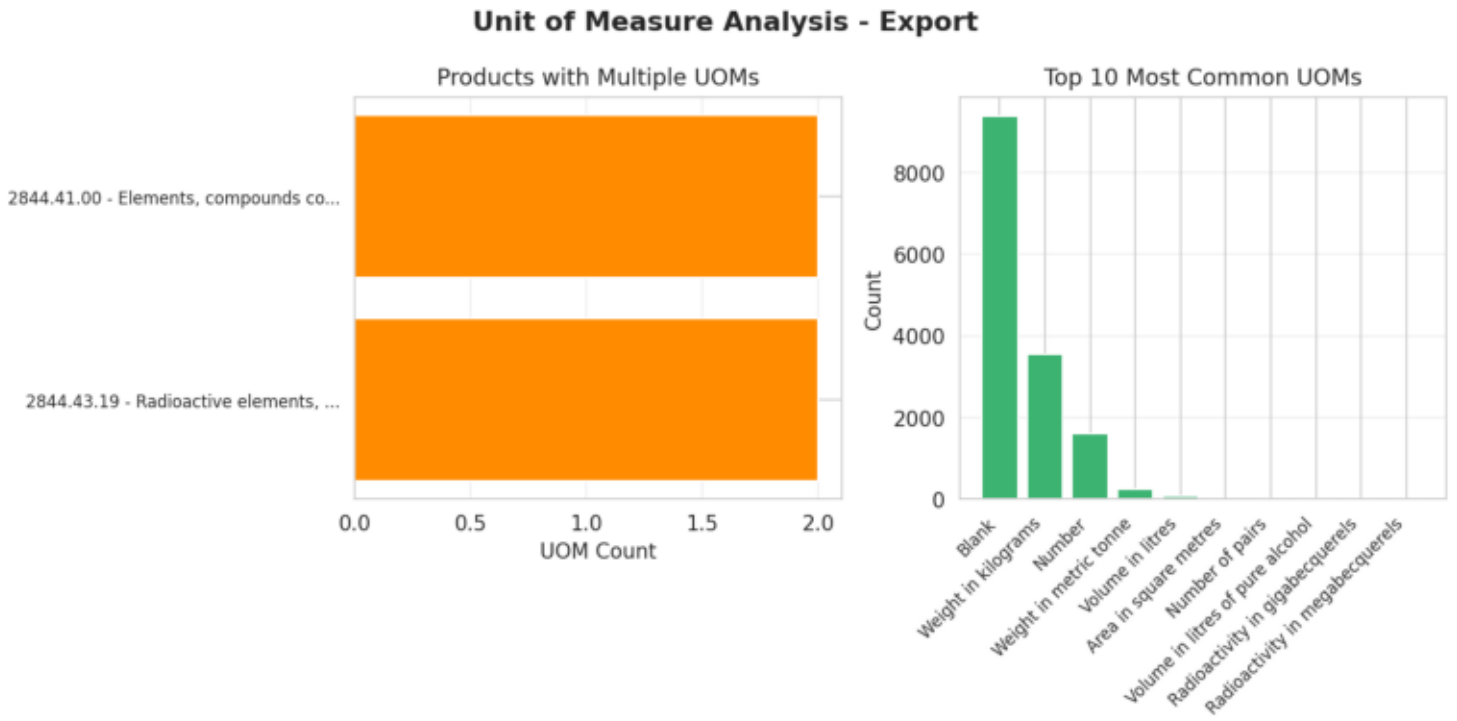

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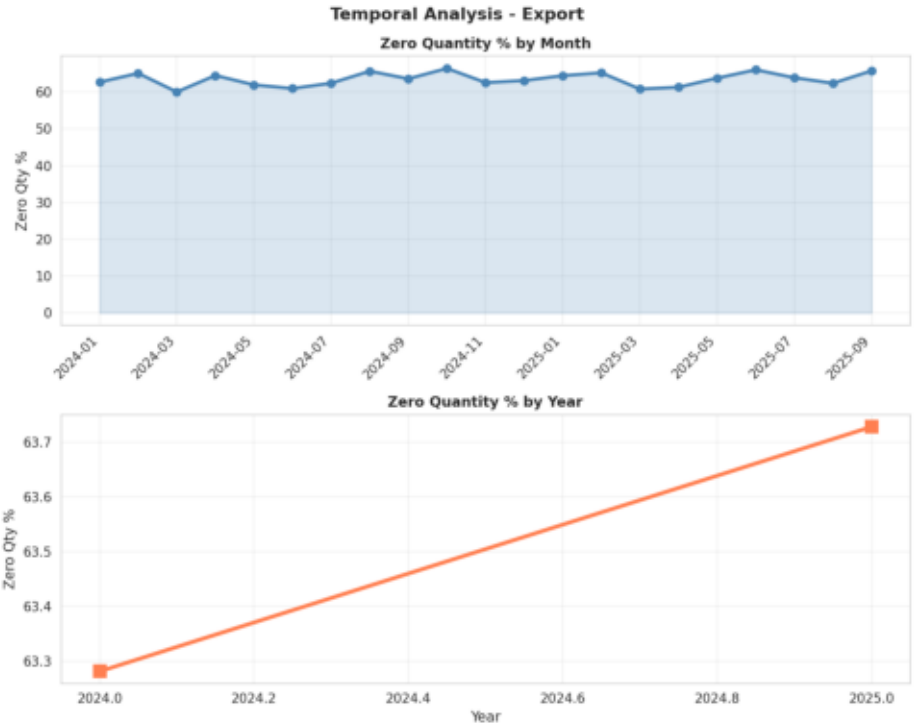

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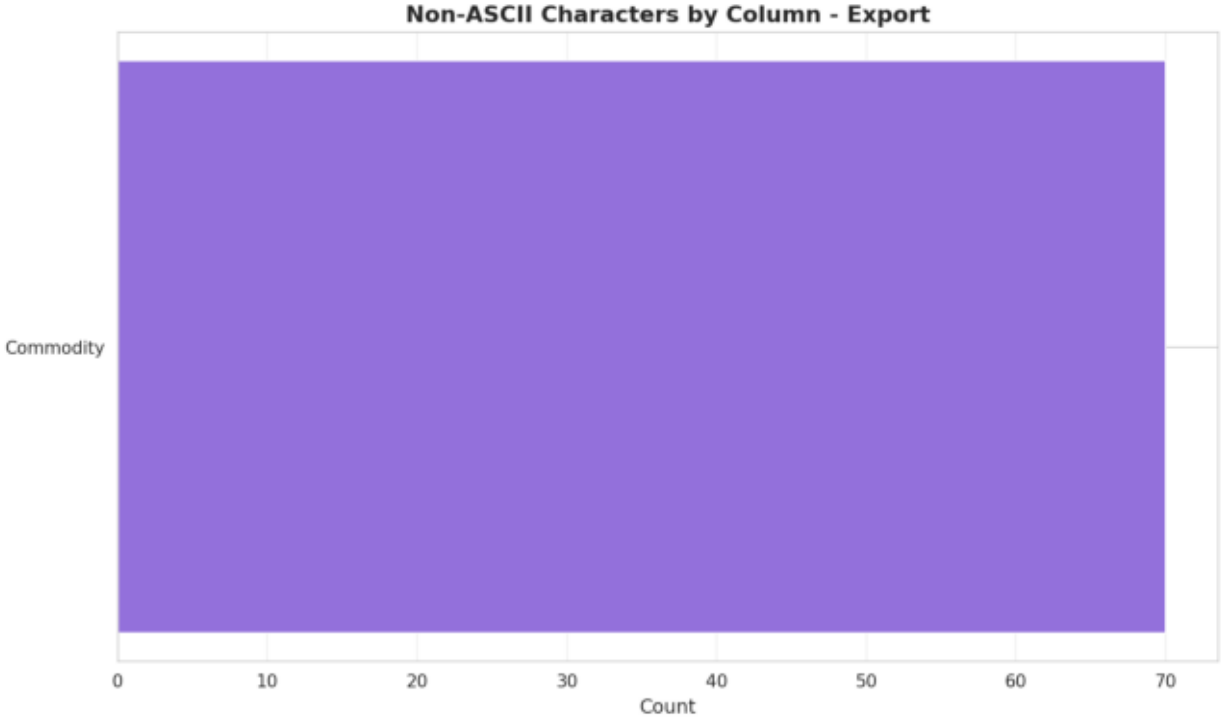

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